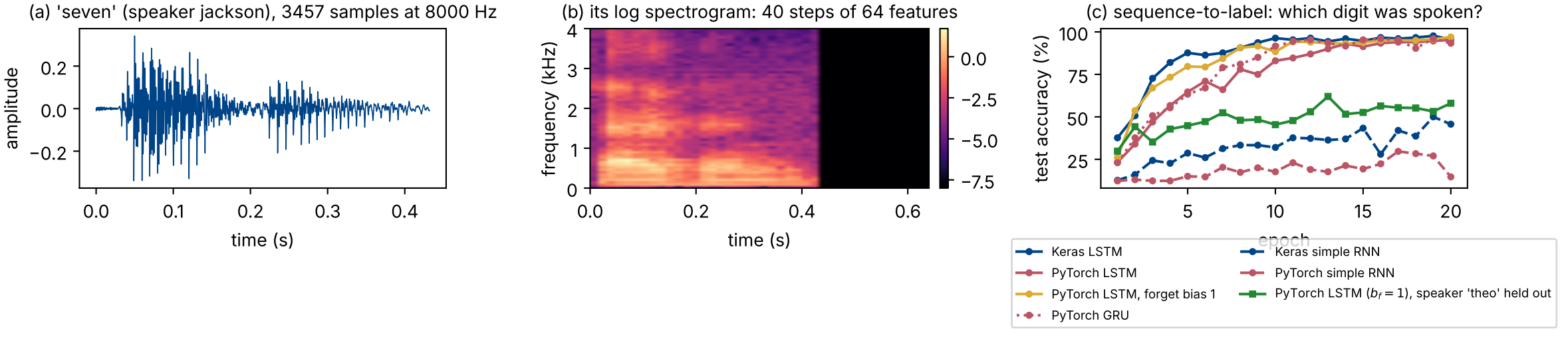

(a) 'seven' (speaker jackson), 3457 samples at 8000 Hz
amplitude
0.2
0.0
−0.2
0.0
0.1
0.2
0.3
0.4
time (s)
(b) its log spectrogram: 40 steps of 64 features
frequency (kHz)
4
3
2
1
0
0.0
0.2
0.4
0.6
time (s)
0.0
−2.5
−5.0
−7.5
(c) sequence-to-label: which digit was spoken?
test accuracy (%)
100
75
50
25
5
10
15
20
epoch
Keras LSTM
PyTorch LSTM
PyTorch LSTM, forget bias 1
PyTorch GRU
Keras simple RNN
PyTorch simple RNN
PyTorch LSTM ($b_f = 1$), speaker 'theo' held out